

PRESSURE TESTING

Manage stored energy, ejected blinds/fittings, leak jets, incorrect isolation and uncontrolled pressurization during piping and equipment pressure tests using an approved test pack and exclusion zone.

PURPOSE AND SCOPE

Manage stored energy, ejected blinds/fittings, leak jets, incorrect isolation and uncontrolled pressurization during piping and equipment pressure tests using an approved test pack and exclusion zone.

KEY HAZARDS

 STORED ENERGY	 EJECTED COMPONENT	 LEAK JET	 WRONG ISOLATION
 OVERPRESSURE	 TRAPPED AIR	 DEAD LEG	 LINE OF FIRE

CRITICAL VALUES TO VERIFY IN THE FIELD

100%

Test-boundary and isolation verification

Calibrated

Correct-range gauge and relief device

0

People inside the pressurized boundary



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

PRESSURE TESTING

Manage stored energy, ejected blinds/fittings, leak jets, incorrect isolation and uncontrolled pressurization during piping and equipment testing.

STANDARDS AND FIELD REQUIREMENTS

ASME B31.3 §345, ASME PCC-2 and the project procedure define test medium, boundary, pressure, gauges, temperature, examination and safe depressurization. The governing design code and site procedure take priority.

ASME B31.3 §345

TEST REQUIREMENTS

Determine the applicable method, pressure, temperature and examination conditions from the design code.

ASME PCC-2 ART. 501

TEST PLANNING

Manage boundary, energy, risk, measurement, safe distance and responsibilities in the written test plan.

OSHA 1910.147

ISOLATION

Reliably isolate connected energy sources and equipment outside the test boundary; verify zero energy.

SAHA TEST PROSEDÜRÜ

PRESSURIZATION AND DEPRESSURIZATION

Pressurize in stages, monitor remotely, never tighten under pressure and depressurize in a controlled manner.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Approved test pack / P&ID | ☐ Test boundary and blind list |
| ☐ Gauge calibrations | ☐ Relief-device data |
| ☐ Test pressure / temperature calculation | ☐ Exclusion-zone and communication plan |

SUPERVISOR DISCUSSION POINTS

- Where does the test boundary start and end?
- How is energy controlled remotely?
- What is the first action for a leak?

HYDROSTATIC TEST, REMOTE MONITORING AND EXCLUSION ZONE



Test pack - Calibrated gauges - Relief - Blinds - Staged pressure - Barrier

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. PREFER HYDROSTATIC TESTING

Use hydrostatic water testing instead of pneumatic testing where practicable and vent trapped air.

2. VERIFY THE TEST BOUNDARY

Independently verify P&ID, blinds, flanges/bolts, supports, gauges and relief device.

3. PRESSURIZE REMOTELY IN STAGES

Increase pressure in stages, read remotely and stop the pump for abnormal behaviour.

4. ISOLATE THE AREA AND FULLY DEPRESSURIZE

Establish a line-of-fire exclusion zone; reduce and verify zero pressure before examination or tightening.

CRITICAL WARNING



Never approach or tighten a pressurized flange, plug, blind, hose or gauge. Stop the test pump and depressurize remotely if leakage occurs.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|---|
| ☐ Test pack approved | ☐ Boundary marked |
| ☐ Blinds verified | ☐ Supports complete |
| ☐ Gauges calibrated | ☐ Relief device suitable |
| ☐ Trapped air vented | ☐ Area isolated |
| ☐ Depressurization route safe | |

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ATTENDANCE AND APPROVAL

Project / Site:

Date:

Presented by:

Work area:

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			